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Techno-economic analysis of large-scale CO_2 ship transport with onboard boil-off gas reliquefaction

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ABSTRACT

This study presents a comprehensive techno-economic analysis of large-scale CO_2 transport by ship over long distances, with a focus on the economic impact of onboard boil-off gas (BOG) reliquefaction. Besides ship size and transport distance, this study includes an extended analysis of the effect of varying seawater temperatures on the reliquefaction system. Various scenarios involving three different ship sizes (40 K, 80 K, 120 K m^3) and transport distances ranging from 1,000 to 20,000 km were examined to evaluate the cost implications for cross-border Carbon Capture and Storage (CCS) projects. The analysis shows that larger ships reduce costs significantly for long distances and large annual CO_2 transport amounts. While the absolute cost of BOG reliquefaction increases with larger ship sizes and longer transport distances, its proportion relative to the total transport cost remains fairly constant at about 5 % of total CO_2 transport costs. Additionally, seawater temperature plays a crucial role in system performance, with higher temperatures significantly increasing cost of reliquefaction system. The study also finds that BOG reliquefaction costs account for approximately 15.1 % to 17.0 % of the ship capital expenditure (CAPEX). Sensitivity analysis identifies the fuel prices and ship CAPEX as the most significant factors influencing overall transport costs. These findings underscore the importance of optimizing ship size and managing fuel costs to design economical CO_2 maritime transport systems, particularly for long-distance CCS applications.

1. Introduction

The global urgency to address climate change has led to increased interest in Carbon Capture and Storage (CCS) technology. CCS is a key strategy for mitigating global warming by capturing carbon dioxide (CO_2) generated from industrial processes and fossil fuel combustion, and safely storing it underground. This technology plays a crucial role in reducing greenhouse gas emissions worldwide and is particularly essential for countries with high energy consumption and limited renewable energy adoption. As the demand for CO_2 reduction grows, CCS technology is gaining more attention, leading to increased research on effective methods for CCS.

For the implementation of CCS, the transport of CO_2 from the capture site to the storage site is inevitable. Various methods for CO_2 transport exist, including pipelines, ships, trains, and tankers, but pipelines and ships are the most commonly used. It is widely recognized that for short distances and large volumes, transporting CO_2 in supercritical or liquid form through pipelines is efficient. However, ships are regarded as more advantageous for longer distances and smaller volumes due to their flexibility, making them ideal for projects involving complex transportation routes. Recently, projects have been proposed to transport CO_2 from multiple capture sites to a central hub using ships, from where it is injected into storage sites. Due to these advantages, interest in CO_2 ship transport is increasing.

In ship transport, CO_2 can be transported under three different conditions: low pressure (0.55 - 0.98 MPa, 218–232 K), medium pressure (1.4 – 2.0 MPa, 243 –253 K), and high pressure (4.5 – 7.2 MPa, 283 - 303 K) (Element Energy, 2018; GASSNOVA, 2016). Numerous studies have been conducted to find the optimal transport conditions, and many recent studies suggest that low-pressure transport near the triple point is the most efficient (Deng et al., 2019; IEAGHG, 2020; Pérez-Bódalo et al., 2023). Low-pressure transport allows for larger tank sizes, increasing the amount of CO_2 that can be transported in a single voyage, and requires less energy for liquefaction, making it more economical (Brownsort, 2015; Engel and Kather, 2018; Kang et al., 2015). Particularly for large amount, long-distance transport, low-pressure conditions

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are considered the most feasible (Roussanaly et al., 2021).

Recently, CCS projects have been scaling up to achieve economies of scale, and cross-border projects are being discussed. These projects are likely to involve the transport of large amount of CO_2 over long distances. However, as mentioned earlier, in the case of large amount ship transport, which is most feasible under low-pressure conditions, boil-off gas (BOG) generation is inevitable due to the low transport temperature. Especially for long-distance transport, the reliquefaction of BOG is essential due to the extended transport time. As the amount of CO_2 transport increases and the transport distance grows, the impact of reliquefaction costs is expected to become more significant. Onboard reliquefaction consumes additional energy beyond engine propulsion, raising overall transport costs (IEAGHG, 2004).

As CCS technology gains recognition as a crucial technology for reducing greenhouse gas emissions, economic analysis studies have also become more active. Studies on the costs of CO_2 transport via pipelines have a long history and have been conducted from various perspectives (Metz et al., 2005; Roussanaly et al., 2014, 2013; Yoo et al., 2013), with researchers now being able to present specific models for pipeline transport costs (Kang et al., 2015). In contrast, studies on the economic analysis of ship transport have been reported relatively recently (Al Baroudi et al., 2021; Aspelund et al., 2006; Element Energy, 2018; IEAGHG, 2020; Weihs et al., 2014). Most studies on ship transport costs have focused on comparing the economics of shipping with pipeline transport based on transport distance and amount (d'Amore et al., 2021; Jung et al., 2013; Kjärstad et al., 2016; Knoope et al., 2015; Nogueira et al., 2022; Roussanaly et al., 2014, 2013; Svensson et al., 2004; ZEP, 2011). More recently, studies have been conducted to determine the most economical shipping conditions, including temperature and pressure (Roussanaly et al., 2021; Seo et al., 2016). While most of the earlier studies focused on short distances within 1000 km or small ships below 20 K (Bjerketvedt et al., 2020; Element Energy, 2018; IEAGHG, 2020; Neele et al., 2017), recent studies have evaluated the economics of ultra-large ships ranging from 50 K to 100 K (d'Amore et al., 2024; Roussanaly et al., 2021; Yuan et al., 2023).

For short-distance transport, the transport duration is short enough that BOG can be managed without reliquefaction by allowing a certain increase in tank pressure. However, for long-distance, large amount transport, reliquefaction of BOG is essential, and reliquefaction systems require compressors and consume additional energy that cannot be overlooked (IEAGHG, 2004), making it necessary to assess their impact on overall transport costs. Most existing studies on ship transport costs focus on short-distance scenarios, largely disregarding the impact of onboard reliquefaction. The impact of reliquefaction has mostly neglected, with only a few studies on CO_2 ship reliquefaction focusing mainly on the design aspects (Lee et al., 2017b). Lee et al. (2017a) compared the energy consumption of various reliquefaction options based on seawater temperature and compressor discharge temperature, suggesting that 60–120 kWh/tCO_2 is required for reliquefaction. Yoo (2017) demonstrated that the use of LNG fuel in CO_2 ships increases reliquefaction efficiency by utilizing the low temperature of LNG. Jeon and Kim (2015) analyzed the impact of CO_2 impurities on reliquefaction during ship transport. Despite these studies, there is a lack of research on the impact of reliquefaction on CO_2 ship transport costs.

This study conducts a techno-economic analysis of long-distance, large-scale CO_2 ship transport, which has recently attracted significant attention. Specifically, the study performs an economic analysis that considers onboard reliquefaction, an aspect often overlooked in the literature, and analyzes its impact on total transport costs. The study evaluates transport costs across 75 scenarios, considering CO_2 transport ships with capacities of 40 K, 80 K, and 120 K m^3, transport distances ranging from 1000 to 20,000 km, and annual transport amount from 1 Mt to 20 Mt. Additionally, a sensitivity analysis was conducted on key variables affecting transport costs.

This paper is structured as follows: Section 2 describes the system boundaries and onboard reliquefaction process considered in the study. Section 3 explains the methodology for the economic analysis. Section 4 presents the results of the economic evaluation, including the impact of reliquefaction, and discusses the results of the sensitivity analysis. Finally, Section 5 provides the conclusions.

2. System overview and reliquefaction modeling

2.1. CO_2 transport chain and system parameters

The target system in this study is a port-to-port CO_2 transport chain, as illustrated in Fig. 1. Liquefied CO_2 (LCO_2) is loaded onto the ship from a storage tank via a loading system and transported by the ship. Upon arrival at the destination port, the CO_2 is unloaded into a storage tank through an unloading system. The storage conditions for CO_2 at the terminal are assumed to be the same as those on the ship, with the temperature, pressure, impurity concentration, and boil-off gas (BOG) rate specified in Table 1. It is assumed that the terminal receives CO_2 at the same temperature and pressure as in the ship's tank, so no liquefaction facility is required at the terminal; however, the reliquefaction of BOG generated at the terminal is considered. Since the focus of this study is the cost analysis of the reliquefaction process in CO_2 transport, the impurity effect (Jeon and Kim, 2015; Vermeulen, 2011; Wang et al., 2011; Wetenhall et al., 2014) which might be important is neglected in this study.

Transport costs were analyzed across 75 scenarios, considering parameters listed in Table 2. The ship sizes considered in this study were 40 K, 80 K, and 120 K m^3. For simplicity, these will be referred to as the 40 K, 80 K, and 120 K ships in this paper.

The 40 K and 80 K ships reflect current trends in shipbuilding, with recent developments indicating their feasibility. In 2023, the largest LCO_2 carrier, with a capacity of 22 K m^3, was contracted, and Approval in Principle (AIP) has been granted for CO_2 transport ships with capacities of 40 K, 70 K, and 74 K m^3 by major shipbuilders. This demonstrates that the construction of 40 K and 80 K ships is both viable and representative of ongoing industry advancements. While the construction of ultra-large 120 K ship presents practical challenges, they were included in the analysis for comparison with the 40 K and 80 K ships to explore potential future developments in CO_2 shipping infrastructure and to evaluate how transport economics would scale with larger ships.

Transport distances ranging from 1000 to 20,000 km were considered, covering most realistic ship transport scenarios. The 10,000 km distance is particularly significant, as it represents the transport distance from Korea or Japan to potential CO_2 storage sites in Australia or the Middle East.

We also considered a range of annual transport capacities, from 1 $MtCO_2$ to 20 $MtCO_2$, to capture the diversity of CCS project scales. This range was chosen to ensure that our analysis addresses both smaller-scale projects as well as large, industrial-scale CCS initiatives, offering a robust examination of the economic implications across different scenarios.

2.2. Boil-off rate modeling

In marine transport, managing BOG (boil-off gas) is crucial, as it is continuously produced during the entire voyage. The handling of BOG becomes especially important during longer voyages, where the accumulated volume can be significant. To prevent the venting of BOG into the atmosphere, reliquefaction systems are essential. These systems help maintain environmental standards and minimize economic losses by efficiently re-liquefying the evaporated gas. Accurate estimation of the Boil-Off Rate (BOR) is crucial for the effective design of these reliquefaction systems, as it directly affects system sizing and efficiency.

For Liquefied Natural Gas (LNG), the BOR has been extensively studied, and systems for managing LNG BOG have been optimized over decades of development (Al Ghafri et al., 2021; Gómez et al., 2013; Hasan et al., 2009; Jeon et al., 2016; Miana et al., 2010; Shin and Lee,

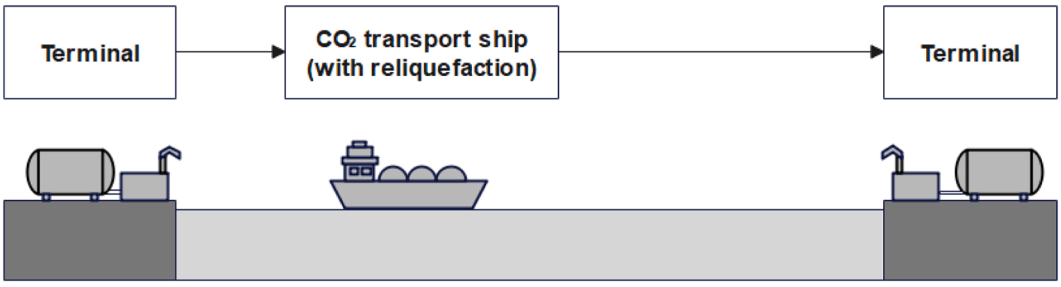


Fig. 1. Port-to-port CO_2 transport supply chain.

Table 1
CO_2 Shipping conditions.

Condition	Value
Temperature (°C)	−50
Pressure (kPa)	700
LCO_2 Density (kg/m^3)	1154
Purity (%)	> 99.99

Table 2
Shipping parameters.

Parameters	Variable
Ship size (m^3, CBM)	40K
	80K
	120 K
Distance (km)	1000
	5000
	10,000
	15,000
	20,000
Annual transport amount (tCO_2)	1M
	2M
	5M
	10M
	20M
Total	75 scenarios

2009; Yin and Ju, 2022). However, for CO_2 tanks used in marine transport for CCS applications, there is limited research available on the BOR (Al Baroudi et al., 2021; Chu et al., 2012; Jeon et al., 2016). This section aims to develop a simplified model to estimate the BOR for CO_2 by comparing it with LNG.

2.2.1. Deriving the simplified boil-off rate equation

To estimate the BOR for a liquid stored in a cryogenic tank, we use a simplified model that calculates heat ingress into the tank using a general thermal resistance model and relates it to the mass of liquid evaporated over a day. The calculations were simplified to extract key physical characteristics of the boil-off process, and the tank was assumed to be spherical for the purpose of this analysis. The BOR is defined as the ratio of the mass of evaporated liquid per day to the total capacity of the tank as given by Eq. (1) (Adom et al. 2010):

$$\mathrm{BOR} = \frac{\dot{m} \times 86400}{M} \times 100 \tag{1}$$

where $\dot{m}$ is the mass of liquid evaporated per second (kg/s), 86,400 is the number of seconds in a day, and M is the total capacity of the tank (kg). The rate of evaporation, $\dot{m}$, is related to the heat ingress rate $\dot{q}$ and the latent heat of vaporization L_v by Eq. (2):

$$\dot{m} = \frac{\dot{q}}{L_v} \tag{2}$$

The total heat ingress rate $\dot{q}$ is determined by the temperature difference between the ambient temperature and the internal temperature of the tank, and the total thermal resistance of the tank as Eq. (3):

$$\dot{q} = \frac{(T_\infty - T_{\mathrm{int}})}{R_{\mathrm{tot}}} \tag{3}$$

where T_∞ is the ambient temperature (K), T_{int} is the internal temperature of the tank (K), and R_{tot} is the total thermal resistance (K/W). The total thermal resistance is given by Eq. (4) (Noh et al., 2019):

$$R_{\mathrm{tot}} = R_{\mathrm{cond},1} + R_{\mathrm{cond},2} + R_{\mathrm{conv}} \tag{4}$$

where $R_{\mathrm{cond},1}$ represents the conduction resistance of the tank wall, $R_{\mathrm{cond},2}$ represents the conduction resistance of the insulation, and R_{conv} represents the convection resistance at the outer surface of the insulation. According to Noh et al. (2019), $R_{\mathrm{cond},2}$ is dominant, and $R_{\mathrm{cond},1}$ and R_{conv} can be considered negligible. Therefore, R_{tot} can be approximated as Eq. (5), assuming that the radius of the tank is much larger compared to the thickness of the insulation:

$$R_{\mathrm{tot}} \approx R_{\mathrm{cond},2} = \frac{1}{4\pi k}\left(\frac{1}{r_2} - \frac{1}{r_3}\right) = \frac{1}{4\pi k}\frac{(r_3 - r_2)}{r_2 r_3} \approx \frac{t}{kA} \tag{5}$$

where k is the thermal conductivity of the insulation (W/m·K), r_2 and r_3 are the inner and outer radii of the insulation layer (m), respectively, t is the thickness of the insulation (m), and A is the surface area of the tank (m^2). Thus, the heat ingress rate ($\dot{q}$) can be expressed as Eq. (6):

$$\dot{q} = \frac{k(T_\infty - T_{\mathrm{int}})\cdot A}{t} \tag{6}$$

The total capacity of the tank is given by Eq. (7):

$$M = \rho \times V \tag{7}$$

where ρ is the density of the liquid (kg/m^3) and V is the volume of the tank (m^3). Substituting the expressions for $\dot{q}$ and M into Eq. (1) gives the BOR equation as Eq. (8):

$$\mathrm{BOR} = \frac{\frac{k(T_\infty - T_{\mathrm{int}})\cdot A}{t\cdot L_v} \times 86400}{\rho \times V} \times 100 \tag{8}$$

Simplifying further, we can get Eq. (9):

$$\mathrm{BOR} = \frac{(T_\infty - T_{\mathrm{int}})\cdot\left(\frac{A}{V}\right) \times 86400}{\left(\frac{t}{k}\right)\cdot\rho\cdot L_v} \times 100 \tag{9}$$

This equation provides the BOR as a function of the thermal conductivity and thickness of the insulation, temperature difference, surface area-to-volume ratio, and product of density and latent heat.

2.2.2. Factors affecting boil-off rate

Based on the derived Eq. (9), we find that four key factors influence the BOR:

- Surface Area-to-Volume Ratio ($\frac{A}{V}$): A lower surface area-to-volume ratio results in a reduced BOR. Therefore, spherical tanks, which inherently have a lower surface area-to-volume ratio compared to cylindrical tanks, are expected to have a lower BOR. Additionally, larger tanks benefit from this effect, making them more efficient in minimizing boil-off rates.
- Temperature Difference ($T_\infty - T_{int}$): A larger temperature difference between the ambient environment and the liquid stored in the tank leads to an increased heat flux, which in turn results in a higher BOR.
- Density and Latent Heat Product ($\rho \cdot L_v$): The product of density and latent heat of vaporization directly affects the BOR. A higher value means more energy is required to vaporize a given volume of liquid, which effectively reduces the BOR.
- Thermal Resistance of Insulation ($\frac{t}{k}$): Higher thermal resistance, achieved through lower thermal conductivity (k) and greater insulation thickness (t), results in better insulation performance, reducing heat ingress and thereby decreasing the BOR.

2.2.3. Application of the boil-off rate model to CO_2 and LNG

In this section, we estimate the BOR for both CO_2 and LNG under specific conditions. The analysis compares the BOR between a standard-sized CO_2 tank (5000 m^3) and LNG tanks of two different volumes: 5000 m^3 and 165,000 m^3. The heat ingress model and BOR equation were applied as outlined in Sections 2.2.1. The input parameters and calculated values used in the BOR estimation, including tank volume, ambient temperature, temperature of the liquid inside the tank, and insulation thickness, are summarized in Table 3.

Below, we present the calculated values for the four key factors affecting BOR, as discussed in Section 2.2.2, along with their impact on BOR. Each factor significantly influences the BOR, and their respective values, including the final BOR, are shown in Table 4.

The calculated BOR values for CO_2 and LNG under different tank sizes provide significant insights into the behavior of cryogenic liquids during marine transport. The smaller LNG tank (5000 m^3) has a BOR of 0.387 % per day, which is higher than the industry standard value of approximately 0.15 % per day. However, the larger LNG tank (165,000 m^3) has a much lower BOR of 0.121 % per day. This difference is primarily attributed to the surface area-to-volume ratio (A/V). Smaller tanks have a higher A/V ratio, resulting in greater relative heat ingress, which increases the BOR. Larger tanks benefit from a lower A/V ratio, thereby reducing heat ingress per unit volume and resulting in a lower BOR.

The BOR for CO_2 in a 5000 m^3 tank is calculated to be 0.124 % per day, which is similar to the BOR for the larger LNG tank (165,000 m^3). Despite the smaller temperature difference (ΔT) for CO_2 compared to LNG, the surface area-to-volume ratio (A/V) heavily influences the BOR, which offsets the effect of the smaller temperature difference. Additionally, the thermal resistance of insulation and the density and latent heat product (ρLv) for CO_2 are different from those of LNG. The lower thermal resistance of insulation and higher latent heat product contribute to reducing the BOR for CO_2.

For LNG, while the calculated BOR for a 165,000 m^3 tank is 0.121 % per day, the industry standard BOR is approximately 0.15 % per day (DNV, 2012). This discrepancy between the calculated and actual values is likely due to the differences between idealized theoretical assumptions and real-world conditions. Practical constraints such as imperfections in insulation, thermal bridging, and variations in tank geometry (which are often not spherical but cylindrical or bi-lobed) contribute to this difference between theoretical models and actual BOR measurements. It is important to note that our modeling is based on an idealized spherical tank, which may not fully capture the complexities of real-world configurations. Similarly, the BOR for CO_2 is also subject to differences between theoretical estimations and real-world values. Therefore, for the purpose of a conservative estimate, this study assumes a BOR of 0.15 % per day for CO_2, analogous to LNG, to account for uncertainties and align with practical considerations in marine transport. This assumption is also applied in the subsequent re-liquefaction cost calculations.

Table 3
Input parameters for boil-off rate calculation.

Parameter	CO_2	LNG	LNG
Tank Volume (m^3)	5000	5000	165,000
Ambient Temperature (K)	308	308	308
Liquid Temperature (K)	224 (at 7 bar)	112 (at 1 atm)	112 (at 1 atm)
Density (kg/m^3)	1152	422	422
Latent Heat of Vaporization (J/kg)	338,681	510,828	510,828
Thermal Conductivity of Insulation (W/mK)	0.047	0.047	0.047
Insulation Thickness (m)	0.2	0.27	0.27
Tank Radius (m)	10.61	10.61	34.03

Table 4
Key factors affecting boil-off rate and their calculated values for CO_2 and LNG.

Key Factor	CO_2	LNG	LNG
Tank Volume (m^3)	5000	5000	165,000
Surface Area-to-Volume Ratio (m^{-1})	0.283	0.283	0.088
Temperature Difference (K)	84.2	196.3	196.3
Density and Latent Heat Product (MJ/m^2)	390.2	215.8	215.8
Thermal Resistance of Insulation (K/W)	4.26	5.74	5.74
Final Boil-Off Rate (%/d)	0.124	0.387	0.121

2.3. Impact of temperature variation on CO_2 reliquefaction characteristics

The reliquefaction process used in this study is a typical two-stage compression-based reliquefaction cycle. Fig. 2 illustrates the thermodynamic pressure-enthalpy (P-H) diagram of the CO_2 reliquefaction cycle along with the corresponding process flow diagram. Fig. 2 shows the P-H conditions when the seawater temperature is 17 °C, which is approximately South Korea's average sea temperature. The approach temperature is 3 °C. The numbered points in the P-H diagram represent key stages before and after each component, indicating changes in temperature, pressure, and enthalpy. The reliquefaction process proceeds as follows:

1) Initial State (①): CO_2 is vaporized from the storage tank at −50 °C and 700 kPa.
2) First Compression (②): The CO_2 is compressed to 3624 kPa in the first compressor, which raises its temperature to approximately 84 °C.
3) Heat Exchange (③): The compressed CO_2 is cooled to 20 °C via heat exchange with seawater.
4) Second Compression (④): The cooled CO_2 is further compressed to 5729 kPa in the second compressor, raising its temperature again to about 60 °C.
5) Heat Exchange (⑤): The recompressed CO_2 is subsequently cooled to 20 °C through a second heat exchanger using seawater.
6) Joule-Thomson (J-T) Cooling (⑥): The pressure is reduced using a J-T valve, resulting in a temperature drop to −50 °C.
7) Separator (⑦): In the separator, CO_2 is divided into liquid and gas phases, achieving a two-phase equilibrium state.

The energy and enthalpy variations at each stage were quantified through numerical calculations using Python. Additionally, an optimization was performed to determine the pressures of the two compressors that would minimize the total work required by both compressors. This optimization employed Powell's method to iteratively adjust the pressure settings of each compressor to achieve the minimum combined work. The thermophysical properties, including enthalpy and entropy at each state point, were determined using the CoolProp library in Python.

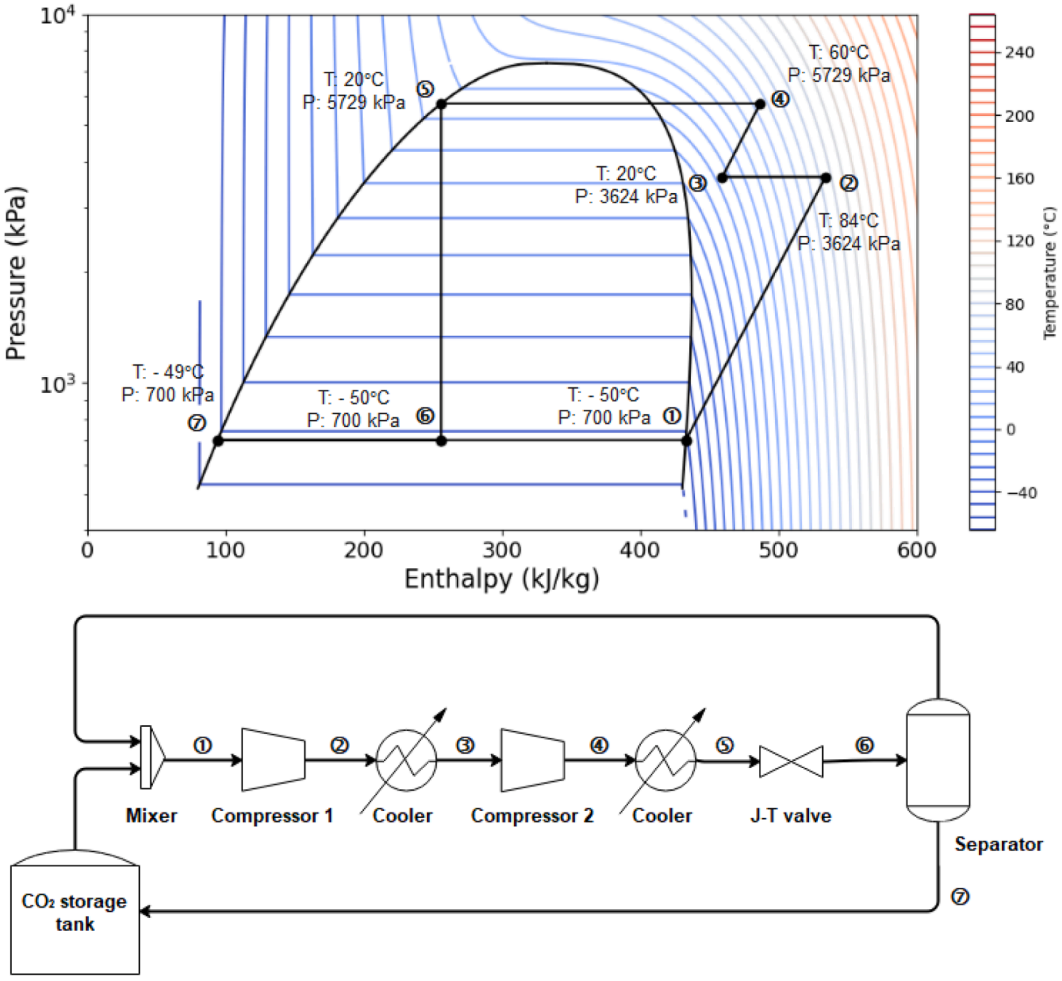


Fig. 2. P-H diagram and process flow diagram for two-stage boil-off gas reliquefaction cycle.

These properties were then used to evaluate the energy requirements and efficiency of the overall process. The capacities of key components and the energy consumption derived from these calculations were used to estimate the capital expenditure (CAPEX) and operational expenditure (OPEX). The capacities of key components, as well as the energy consumption derived from these calculations, formed the basis for further analysis in Chapter 4.

3. Techno-economic analysis methodology

This section explains the methodology used to conduct the techno-economic analysis for the 75 transport scenarios outlined in Table 2. Costs were calculated for the transport chain depicted in Fig. 1, considering the CAPEX and OPEX of the ships, reliquefaction systems, and terminals separately. The analysis included the cost of the ship returning empty after delivering CO_2 to the destination terminal. For the terminals, both the loading and unloading terminals were considered, including their respective CAPEX and OPEX. The key transport parameters needed for the cost calculations are presented in Table 5.

The transportation cost per ton of CO_2 for each scenario was calculated using the following Eq. (10):

$$CO_2 \text{ transport cost } (US\$ / tCO_2) = \frac{CAPEX_a + OPEX}{\text{Amount of annual } CO_2 \text{ transported}} \quad (10)$$

Here, $CAPEX_a$ is the annualized CAPEX, calculated by applying the equivalent annual annuity method to the CAPEX (Newnan et al., 2004), using Eq. (11) below.

Table 5
Economic analysis parameters and basic assumptions.

Parameters	Symbol	Value
Ship speed (knot)	S_{ship}	15
Loading time (h)	t_{loding}	12
Unloading time (h)	$t_{unloading}$	12
Approach & departure time (h)	$t_{a\&d}$	12
Percentage of tank filling		
Outbound (%)	η_{out}	98
Return (%)	η_{ret}	5
Fuel consumption rate (kg/h/1000 m^3) [1)]	F	
40 K ship	F_{40K}	35.8
80 K ship	$F_{80\ K}$	25.1
120 K ship	F_{120K}	21.6
Price of fuel marine diesel oil (MDO) (US$/kg)	P_f	0.635
Price of electricity (US$/kWh)	P_e	0.0769
Terminal storage tank capacity compared to ship volume (%)	C_{st}	150

[1)] The value is based on regressions established by Element Energy (2018).

$$CAPEX_a = \frac{r \times CAPEX}{1 - (1+r)^{-n}} \quad (11)$$

Where r is the discount rate used to calculate the present value of future cash flows, assumed to be 7 %. The project lifetime, n, is considered to be 25 years. OPEX is the sum of two components: the fixed OPEX (fOPEX), which covers annual operating costs such as maintenance, labor, and insurance, and the variable OPEX (vOPEX), which accounts for the annual costs of fuel and electricity consumed as shown in Eq. (12):

$$OPEX = fOPEX + vOPEX \quad (12)$$

The fOPEX was estimated to be 5 % of the CAPEX.

3.1. CAPEX and OPEX for ship

3.1.1. CAPEX of ship

Table 6 shows the per-ship CAPEX for the three sizes of LCO_2 transport ships used in this study. The per-ship CAPEX of the 40 K ship was estimated based on the contract price reported for a 22 K LCO_2 transport ship and the contract prices for three LPG ships with similar transport conditions to LCO_2 ships (two 40 K and one 45 K LPG ships). This per-ship CAPEX for the 40 K ship, selected as the reference capacity, was then used to estimate the per-ship CAPEX for the 80 K and 120 K ships using the scaling law shown in Eq. (13).

$$\text{Ship CAPEX} = \text{Reference investment cost} \cdot \left(\frac{\text{Considered capaticy}}{\text{Reference capacity}}\right)^{\alpha} \quad (13)$$

Where α is the scale coefficient, which varies depending on the equipment applied (Peters et al., 2003). Based on a regression analysis of ship cost variations from reference (Veldman, 2011), the value of α was set at 0.75 in this study. The estimated per-ship CAPEX for the 40 K, 80 K, and 120 K ships is shown in Table 6, and a comparison with other literature is presented in Fig. 3. The per-ship CAPEX for the 40 K ship was estimated based on actual contract prices. The per-ship CAPEX estimated for the 80 K and 120 K ships using the scaling law also falls within a similar range to those reported in other studies as shown in Fig. 3.

To calculate the total ship CAPEX required to achieve the annual transport amount, it is necessary to determine not only the price per ship but also the total number of ships needed. To determine the number of ships required, calculating the number of annual round trips is a prerequisite. This can be derived from the CO_2 mass transported in a single round trip and the time required for that trip. First, the mass of CO_2 that a single ship can transport in one round trip, Q_t, is calculated using Eq. (14).

$$Qt = Qout - - Qret = V_{ship} \cdot \eta_{out} \times CO_2\ density\ V_{ship} \cdot \eta_{ret} \times CO_2\ density \quad (14)$$

Where Q_{out} is the outbound CO_2 transport mass and Q_{ret} is the returned CO_2 transport mass. Q_{ret} must be considered because a portion of CO_2 needs to be transported on the return trip to maintain the temperature and pressure of the CO_2 tanks, rather than returning with an empty ship. V_{ship} is the ship capacity in cubic meters. η_{out} and η_{ret} are the percentages of tank filling for outbound and return transports, respectively. The CO_2 density used in this calculation is 1.1546 kg/m^3. The time required for one round trip is determined using Eq. (15) and Eq. (16).

$$T_t = t_{shipping} + t_{loding} + t_{unloading} + 2 \cdot t_{a\&d} \quad (15)$$

$$t_{shipping} = \frac{2 \cdot D}{S_{ship}} \quad (16)$$

Where T_t represents the total time required for a round trip. $t_{shipping}$, t_{loding}, $t_{unloading}$, and $t_{a\&d}$ are the times for shipping, loading, unloading, and the approach and departure time, respectively. D is the distance between ports, and S_{ship} is the ship's speed. Based on the round-trip time, the number of annual round trips is calculated using Eq. (17).

Table 6
Per-ship CAPEX.

Ship size (m^3)	Per-Ship CAPEX (M US$/ship)
40K	55.6
80K	93.6
120K	126.8

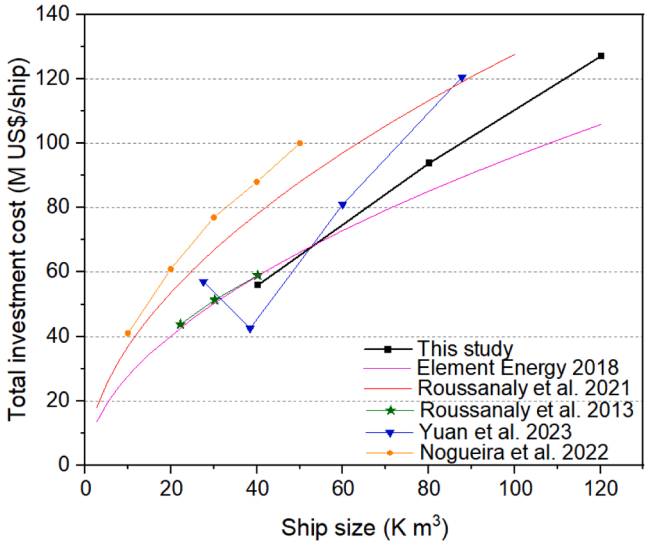


Fig. 3. Comparison of this study's per-ship CAPEX with references' costs.

$$N_{ar} = \left\lfloor \frac{8760 \cdot \eta_{shipping}}{T_t} \right\rfloor \quad (17)$$

Where N_{ar} is the number of annual round trips. $\eta_{shipping}$ is the shipping efficiency, accounting for operational limits due to adverse weather conditions and other factors. In this study, it is assumed that a ship can operate for a maximum of 8000 h per year. Finally, the number of ships required to meet the annual transport amount is calculated using Eq. (18).

$$N_{ship} = \left\lceil \frac{Q_a}{Q_t \cdot N_{ar}} \right\rceil \quad (18)$$

Where N_{ship} is the number of ships required, and Q_a is the annual transport amount for CO_2. Finally, this leads to the total ship CAPEX for the CO_2 transportation target and ship capacity being calculated using Eq. (19).

$$CAPEX_{ship} = C_{ship} \cdot N_{ship} \quad (19)$$

Where $CAPEX_{ship}$ represents the total ship CAPEX required for the scenarios, and C_{ship} is the per-ship CAPEX. The number of ships required for each of the 75 scenarios is presented in Table 7.

Table 7
Number of ships required.

Ship size (m^3)	Annual amount (tCO_2)	Distance (km)				
		1000	5000	10,000	15,000	20,000
40K	1M	1	2	3	4	5
	2M	1	3	5	7	10
	5M	2	7	12	17	24
	10M	4	13	24	34	47
	20M	8	25	47	67	94
80K	1M	1	1	2	2	3
	2M	1	2	3	4	5
	5M	1	4	6	9	12
	10M	2	7	12	17	24
	20M	4	13	24	34	47
120K	1M	1	1	1	2	2
	2M	1	1	2	3	4
	5M	1	3	4	6	8
	10M	2	5	8	12	16
	20M	3	9	16	23	32

3.1.2. OPEX of ship

As mentioned in Eq. (3), the operating expenses (OPEX) consist of the sum of fOPEX and vOPEX. The fOPEX was assumed to be 5 % of the total ship CAPEX. The vOPEX only considered the fuel costs for the ships used in annual transportation, and these costs were calculated using Eq. (20).

$$\text{vOPEX} = F \cdot T_t \cdot N_{ship} \cdot N'_{ar} \quad (20)$$

Where F is the fuel consumption rate. Marine Diesel Oil (MDO) was used as the ship fuel, and the fuel consumption rates for each ship are shown in Table 5.

3.2. CAPEX and OPEX for reliquefaction

In this study, the CO_2 reliquefaction system was simulated using Python, and based on this, the capacity and energy consumption of each piece of equipment were calculated. These derived values were then used to estimate the CAPEX and OPEX for reliquefaction systems on ships with capacities of 40 K, 80 K, and 120 K m^3.

The key components of the reliquefaction system are compressors and heat exchangers. The capacities of compressor 1, compressor 2, heat exchanger 1, and heat exchanger 2 required for reliquefaction on 40 K, 80 K, and 120 K ships are listed in Table 8.

The CAPEX for these components was calculated using the Scaling Law, similar to the method used for the ships. The reference costs for each piece of equipment were obtained from Guthrie (1969) and Yoo (2017). The scaling exponents used were 0.77 for compressors and 0.65 for heat exchangers (Peters et al. 2003). Additionally, the Chemical Engineering Plant Cost Index (CEPCI) was applied to update the equipment costs to current prices

To ensure accurate cost estimation, the concept of Bare Module Cost (BMC) was introduced to calculate the costs of individual equipment. The BMC includes the cost of the equipment itself, along with all necessary auxiliary components and installation costs. The BMC was calculated by multiplying the equipment cost (E) by the Bare Module Factor (B) and the number of replacements (R). The Bare Module Factor for compressors was set at 2.15, and for heat exchangers, it was set at 3.17 (Ulrich and Vasude 2004). The number of replacements was set at two for compressors and one for heat exchangers.

The OPEX, similar to that for the ships, consists of fOPEX and vOPEX. The fOPEX was estimated at 5 % of the CAPEX, while the vOPEX was calculated based on the annual operating hours and fuel consumption. The annual fuel consumption for vOPEX was calculated by multiplying the annual energy consumption by the lower heating value (LHV) of the fuel. The annual energy consumption was determined by dividing the energy consumption of the compressors by the generator efficiency, then multiplying by the annual operating hours. The key parameters used for estimating vOPEX were an annual operating time of 8000 h, a generator efficiency of 40 %, a fuel price of $0.635/kg, and an LHV of 0.084 kg/kWh. The fuel used for reliquefaction was the same as that for the ships, MDO. The final CAPEX and OPEX estimates for the onboard reliquefaction system are presented in Table 9.

Table 8
Machine capacities (kW) for key equipment for different ship sizes.

Ship size (m^3)	Equipment	Machine capacity (kW)
40K	Compressor 1	312.6
	Compressor 2	128.9
	Heat Exchanger 1	170.0
	Heat Exchanger 2	427.2
80K	Compressor 1	625.2
	Compressor 2	257.8
	Heat Exchanger 1	340.1
	Heat Exchanger 2	854.4
120K	Compressor 1	937.8
	Compressor 2	386.7
	Heat Exchanger 1	510.1
	Heat Exchanger 2	1281.7

3.3. CAPEX and OPEX for terminal

In this section, the CAPEX and OPEX for CO_2 terminals designed to handle 40 K, 80 K, and 120 K ships were estimated. It is important to note that the CAPEX of a CO_2 terminal is primarily determined by the capacity of a single CO_2 transport ship rather than the annual transport amount. This is because, in terms of CAPEX, the terminal only needs to be equipped with the capacity required for the loading and unloading of a single ship. Regarding OPEX, differences in annual transport amount affect the number of loading/unloading operations. While these differences could impact the OPEX of pumps used in loading/unloading, this effect is minimal and was therefore disregarded in this study.

To estimate the CAPEX for the terminal, the BMC was first calculated based on the individual equipment costs. The bare modules considered for CO_2 terminal CAPEX include CO_2 tanks, CO_2 loading/unloading arms, and the reliquefaction system. The total BMC was then adjusted by adding a 20 % contingency to derive the Total Direct Cost. To this, Indirect, Owner, and Project Contingency costs were added to calculate the final CAPEX. The Indirect, Owner, and Project Contingency costs were assumed to be 41 % of the Total Direct Cost.

The total CO_2 tank capacity required at the terminal was assumed to be 1.5 times the ship capacity. The capacity of a single CO_2 tank was fixed at 5000 m^3. The BMC for a single CO_2 tank was set at 3650,000 USD. The BMC for the CO_2 loading/unloading arm was calculated assuming a cost of 1.8 USD per ton of CO_2 handled (Element Energy 2018). The CAPEX for the reliquefaction system at the CO_2 terminal was assumed to be the same as the reliquefaction system on the corresponding ship.

The OPEX for the CO_2 terminal, like that for the ships, consists of fOPEX and vOPEX. The fOPEX was estimated at 5 % of the CAPEX, while the vOPEX included reliquefaction costs and other energy expenses. The amount of electricity required for reliquefaction at the CO_2 terminal was assumed to be the same as that required for the reliquefaction system on the corresponding CO_2 ship. The electricity rate was set at 0.0769 USD/kWh. While the ship generates electricity onboard, consuming fuel as part of its OPEX, the terminal receives electricity directly from the shore, so the electricity cost was accounted for in the terminal's OPEX. At the terminal, any additional energy requirements, aside from those needed for reliquefaction, were assumed to be equal in amount to the

Table 9
Estimation of CAPEX and OPEX for reliquefaction.

Ship size (m^3)	CAPEX		OPEX	
	Cost item	Estimate (K US $)	Cost item	Estimate (K US $)
40K	Compressor 1	569	fOPEX	117
	Compressor 2	288	vOPEX	276
	Heat exchanger 1	18.9		
	Heat exchanger 2	16.9		
	Total	3796	**Total**	393
80K	Compressor 1	970	fOPEX	199
	Compressor 2	490	vOPEX	551
	Heat exchanger 1	29.7		
	Heat exchanger 2	26.5		
	Total	6457	**Total**	751
120K	Compressor 1	1325	fOPEX	271
	Compressor 2	670	vOPEX	827
	Heat exchanger 1	38.7		
	Heat exchanger 2	34.5		
	Total	8812	**Total**	1099

reliquefaction energy.

The final CAPEX and OPEX estimates for the CO_2 terminal are presented in Table 10.

4. Results and discussion

4.1. Analysis of CO_2 reliquefaction costs

This section presents a detailed analysis of the CAPEX and OPEX associated with the CO_2 reliquefaction process. The study aims to evaluate how varying environmental conditions, particularly changes in seawater temperature, impact the economic feasibility of the reliquefaction process.

The technical performance analysis was conducted for a typical two-stage compression-based CO_2 reliquefaction cycle, as described in Section 2.3. The CAPEX and OPEX calculations were performed using the methodologies detailed in Section 3.2. The analysis is divided into two subsections: Section 4.1.1 focuses on evaluating the thermodynamic performance of the system for a 40 K ship, while Section 4.1.2 provides a comparative analysis of CAPEX and OPEX for different ship sizes under varying seawater temperatures.

4.1.1. Effect of temperature on system performance for a 40K ship

This subsection analyzes how varying seawater temperatures affect the thermodynamic characteristics of the CO_2 reliquefaction process for a 40 K ship, including pressure conditions, vapor ratios, and mass flow rates. The ultimate goal is to determine the compressor work required under different operational conditions, which is crucial for assessing system efficiency and economic feasibility.

The temperature and pressure at Points 2 and 4 in the reliquefaction cycle play significant roles in determining system efficiency. Fig. 4 show the variation in temperatures and pressures at Points 2 and 4 for discharge temperature limits of 100 °C, 110 °C, and 120 °C. As the seawater temperature rises from 15 °C to 35 °C, the temperatures and pressures at Points 2 and 4 increase until the discharge temperature limit is reached. Once the limit, particularly at 100 °C, is reached, a bottleneck occurs in the system. This condition restricts the ability of the compressor to further increase the pressure, leading to a reduction in pressure at Point 4. This phenomenon occurs because the compression process is isentropic. As the seawater temperature rises, the cooled CO_2 temperature at Point 3 also increases before entering the second compressor. With this elevated temperature, the isentropic compression process results in a lower final pressure when reaching the discharge temperature limit, causing the pressure at Point 4 to drop. For instance, at the 100 °C discharge limit, the system reaches this temperature when the seawater temperature is around 30 °C. Beyond this point, as the seawater temperature continues to increase, the discharge pressure of the compressor decreases. A similar effect is observed at the 110 °C limit, where the seawater temperature of 34 °C leads to a drop in pressure by the time it reaches 35 °C.

The vapor ratio at the separator is a key parameter indicating the fraction of vapor present in the two-phase mixture, which directly influences the mass flow through the system. A higher vapor ratio means that more CO_2 remains in the gaseous state, requiring additional work for reliquefaction. As seawater temperature increases, the vapor ratio also rises due to the reduced cooling capacity of the system, ultimately increasing the overall mass flow rate. This phenomenon can be described using a geometric progression concept, where the total mass flow rate at the separator becomes $1/(1-x)$ given a vapor fraction of x. As the vapor ratio approaches 1, the mass flow rate increases sharply, adding significant load to the compressors. This behavior is particularly noticeable when the discharge temperature limit is reached early, as the vapor fraction tends to increase significantly, resulting in a considerable increase in mass flow, which must be managed by the compressors.

Fig. 5 illustrates the relationship between the vapor ratio and the overall mass flow rate as the seawater temperature increases from 15 °C to 35 °C for different discharge temperature limits (100 °C, 110 °C, and 120 °C). As the seawater temperature rises, the vapor ratio steadily increases, which subsequently leads to a rise in the mass flow rate through the system. The vapor ratio at the separator increases significantly as the system reaches its discharge temperature limit prematurely, especially at the 100 °C limit. This effect is more pronounced at higher seawater temperatures. For instance, at a seawater temperature of 35 °C and a discharge temperature limit of 100 °C, the vapor ratio reaches approximately 0.82, resulting in a sharp rise in the mass flow rate, which becomes approximately 3.04 times greater than the value at a seawater temperature of 15 °C. This occurs because reaching the discharge limit prevents Compressor 2 from sufficiently increasing the pressure, leading to a considerable increase in the vapor ratio. In contrast, for discharge temperature limits of 110 °C and 120 °C, the vapor ratios and corresponding mass flow rates show a comparatively gradual increase, highlighting the importance of selecting discharge temperature limits to control vapor ratios and overall mass flow rate effectively.

As seawater temperature increases, the compressor work for both Stage 1 and Stage 2 also increases consistently across all discharge temperature limits, as shown in Fig. 6. The primary reason for this trend is that compressor work is directly related to the product of mass flow and the enthalpy difference between the inlet and outlet.

Based on the earlier discussion about mass flow, it is evident that as the vapor ratio increases with higher seawater temperatures, the mass flow rate through the system also increases. The enthalpy difference between the inlet and outlet of the compressors also grows as the temperature and pressure increase. Since higher seawater temperatures lead to increased temperatures at Points 2 and 4 in the CO_2 reliquefaction cycle, the enthalpy at each point becomes higher, which directly results in a greater energy requirement for compression.

The performance of the CO_2 reliquefaction system is heavily dependent on the effectiveness of the heat exchanger, which plays a crucial role in maintaining cooling efficiency. Variations in seawater temperature significantly impact the operating conditions of the heat exchanger, necessitating an assessment of the required capacity to maintain optimal system performance under different thermal conditions. Specifically, when the heat exchanger is subjected to seawater at elevated temperatures, it no longer functions effectively as a condenser but instead behaves as a cooler, causing the CO_2 to remain in a supercritical state rather than condensing into a liquid. In the two-phase region, heat transfer is significantly enhanced due to the high heat transfer coefficient associated with phase change. However, in the supercritical region, this enhancement disappears, potentially leading to a lower overall heat transfer rate. This raises concerns about whether a considerably larger heat transfer area is necessary to achieve the same cooling performance under supercritical conditions.

This study examines how supercritical cooling influences the required heat transfer area compared to phase change conditions. To

Table 10
Estimation of CAPEX and OPEX for terminal.

Ship size (m^3)	CAPEX		OPEX	
	Cost item	Estimate (K US$)	Cost item	Estimate (K US$)
40K	Total Direct Cost	57,214	fOPEX	2750
	Indirect+Owner+Project Contingency	23,458	vOPEX	792
	Total	80,671	**Total**	3542
80K	Total Direct Cost	113,067	fOPEX	5465
	Indirect+Owner+Project Contingency	46,357	vOPEX	1548
	Total	159,424	**Total**	7049
120K	Total Direct Cost	168,551	fOPEX	8170
	Indirect+Owner+Project Contingency	69,106	vOPEX	2376
	Total	237,657	**Total**	10,547

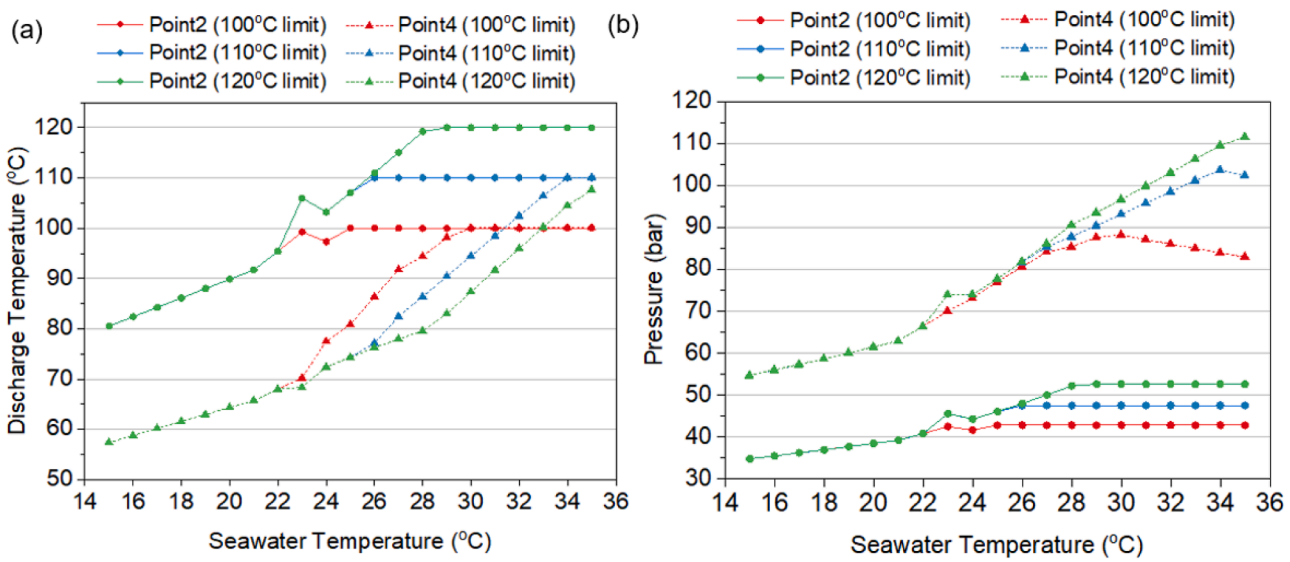


Fig. 4. Effect of Seawater Temperature on the CO_2 Reliquefaction Cycle at Different Compressor Discharge Temperature Limits: (a) Temperature Changes, (b) Pressure Changes at Points 2 and 4.

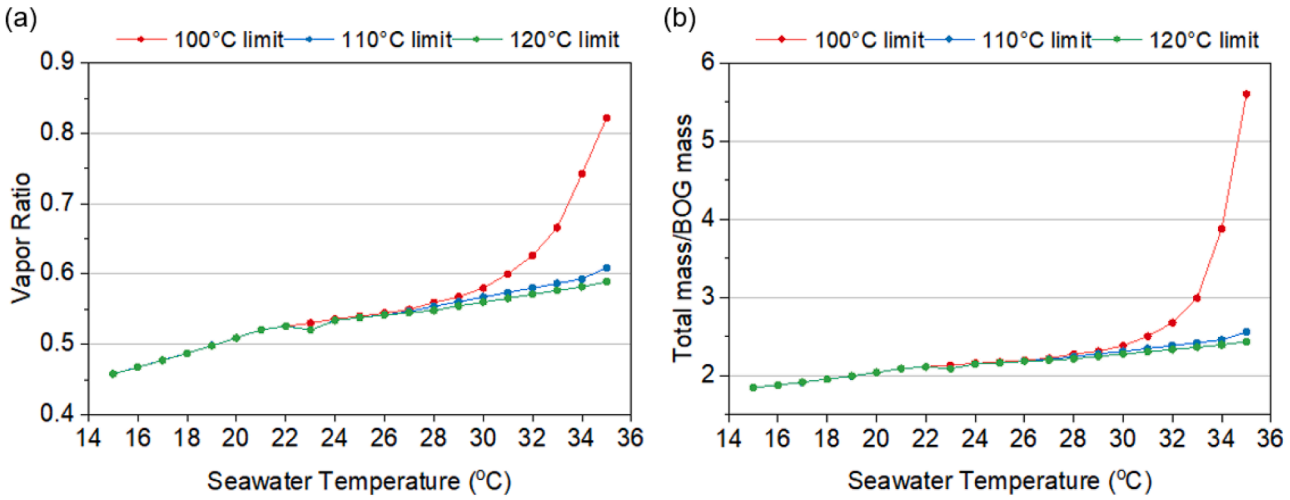


Fig. 5. Effect of seawater temperature on (a) vapor ratio and (b) total mass/BOG mass.

analyze this effect, two cases are considered. In Case 1, gaseous CO_2 undergoes a phase change and transitions into a liquid state, while Case 2 investigates cooling within the supercritical region where phase change does not occur. For both cases, a shell-and-tube heat exchanger with a counter-flow configuration consisting of fifty CO_2 tubes with a diameter of 0.025 m is assumed. The minimum approach temperature is set to 3 °C to provide sufficient thermal driving force for heat exchange.

The heat transfer area is computed using the Log Mean Temperature Difference (LMTD) method (Incropera et al., 2007; Kern, 1950) and energy balance equations, as shown in Eq. (21) below:

$$A = \frac{Q}{U \times \text{LMTD}} \tag{21}$$

where A is the heat transfer area, Q is the heat transfer rate required by the reliquefaction system to cool CO_2 and facilitate its phase transition, and U is the overall heat transfer coefficient. To determine U, the following Eq. (22) is used:

$$\frac{1}{U} = \frac{1}{h_{\text{hot}}} + R_{\text{wall}} + \frac{1}{h_{\text{cold}}} \tag{22}$$

Each term on the right-hand side represents a specific thermal resistance. h_{hot} is the convective heat transfer coefficient for the hot fluid, which corresponds to the tube side in a shell-and-tube heat exchanger, while and h_{cold} is the convective heat transfer coefficient for the cold fluid, corresponding to the shell side For single-phase regions, including supercritical conditions, h_{hot} is determined using the Dittus-Boelter equation (Incropera et al., 2007). In the two-phase region, the Shah correlation (Shah, 1979) is applied. R_{wall}, assumed to be 0.000125 m^2K/W , represents the thermal resistance of the heat exchanger wall, while h_{cold}, taken as 400 W/m^2K, is a typical shell-side heat transfer coefficient in shell-and-tube heat exchangers.

The LMTD is given by Eq. (23):

$$\text{LMTD} = \frac{\Delta T_{\text{in}} - \Delta T_{\text{out}}}{\ln(\Delta T_{\text{in}}/\Delta T_{\text{out}})} \tag{23}$$

where ΔT_{in} is the temperature difference between the hot fluid entry and the cold fluid exit, and ΔT_{out} is the temperature difference between the hot fluid exit and the cold fluid entry.

A simplified diagram of the counter-flow heat exchanger, incorporating both phase-change and supercritical cooling cases, is shown in

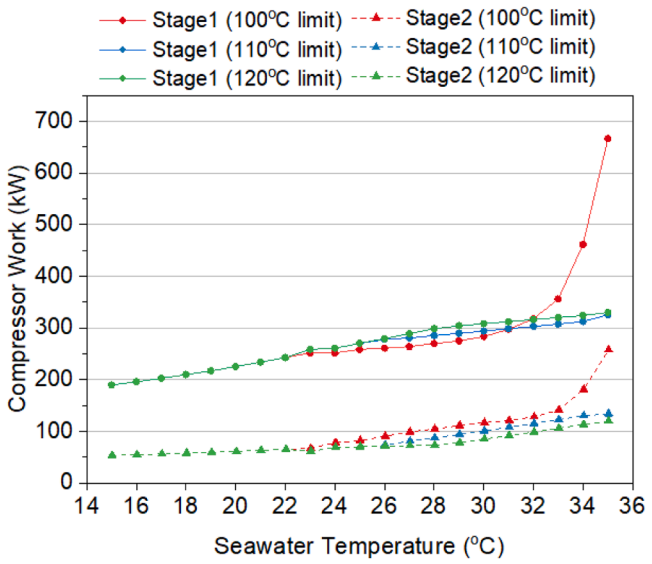


Fig. 6. Effect of seawater temperature on the compressor work.

Fig. 7. In Case 1, Eq.(21) ~ Eq. (23) were applied separately to gas region, two-phase region, and liquid region. The summarized results of the calculations are presented in Table 11, while Fig. 8(a) and Fig. 8(b) illustrate the temperature variations along the heat exchanger for Case 1 and Case 2, respectively.

In Case 1, gaseous CO_2 is cooled from 77.4 °C to 27.0 °C using seawater at 24 °C through gas, phase-change, and liquid regions, as shown in Fig. 8(a). The overall heat transfer coefficients, calculated using Eq. (22), are 157 W/m²·K for the gas region, 304 W/m²·K for the two-phase region, and 225 W/m²·K for the liquid region, which requires a total heat transfer area of 4.61 m². In this scenario, although the two-phase region exhibits a significantly higher heat transfer coefficient, its relatively small temperature difference limits the overall impact of that enhanced heat transfer. In contrast, the gas region in Case 1 maintains a larger temperature difference but with a relatively low heat transfer coefficient. In Case 2, CO_2 is cooled from 100.0 °C to 38.0 °C using seawater at 35 °C without phase change, as shown in Fig. 8(b). The overall heat transfer coefficient is 226 W/m²·K, which requires a total area of 3.45 m². Although the phase-change region in Case 1 provides a noticeably higher heat transfer rate, the large temperature difference in Case 2, as shown in Fig. 8(b), compensates for the absence of phase change. Consequently, even though phase change enables a notably higher heat transfer rate, Case 2 can maintain comparable cooling performance with a smaller total heat transfer area.

4.1.2. CAPEX and OPEX analysis by temperature

This section presents an analysis of the impact of seawater temperature variations on the CAPEX and OPEX for CO_2 reliquefaction systems for three different ship sizes—40 K, 80 K, and 120K—considering seawater temperatures ranging from 15 °C to 35 °C. This analysis particularly focuses on understanding how costs change under typical annual conditions, such as an average seawater temperature of 17 °C (the average seawater temperature in South Korea), compared to peak summer conditions, such as 34 °C, which represents a high seawater temperature with a 2 °C margin above the highest recorded temperature.

At a seawater temperature of 17 °C, CAPEX and OPEX for the 40 K, 80 K, and 120 K ships range from approximately $2.49 million and $0.35 million to $5.76 million and $1.01 million, respectively. When the seawater temperature increases to 34 °C, these costs rise significantly—CAPEX increases by approximately 1.6 times, and OPEX increases by around 1.7 times for all ship sizes, as illustrated in Fig. 9. This rise highlights the significant effect of increased seawater temperature on cost due to reduced system efficiency.

The key driver for this increase in costs is the increased workload on the compressors. As discussed in Section 4.1.1, higher seawater temperatures reduce the system's cooling efficiency, leading to a higher vapor ratio in the separator and a greater mass flow rate. This, in turn, directly increases the work required by the compressors. As seawater

Table 11
Summary of the parameters and results for the heat exchanger calculation.

Parameter		Case 1	Case 2
Seawater Temperature (°C)		24	35
CO_2 Inlet Temperature (°C)		77.4	100.0
Pressure (Pa)		7.30×10^6	8.29×10^6
CO_2 Mass Flow Rate (kg/s)		1.69	4.40
CO_2 Density (kg/m³)		176.43	184.77
CO_2 Viscosity (Pa·s)		19.22×10^6	20.24×10^6
CO_2 Specific Heat Capacity (J/kg·K)		1886.82	1746.68
CO_2 Thermal Conductivity (W/m·K)		0.03	0.03
Seawater Specific Heat Capacity (J/kg·K)		4160.89	4170.29
Heat Transfer Rate of Heat Exchanger (W)		4675.02	12,772.44
Heat Transfer Coefficient (W/m²·K)	Gas Region	267.38	557.47
	Two-Phase Region	1525.01	
	Liquid Region	551.51	
Overall Heat Transfer Coefficient (W/m²·K)	Gas Region	157.11	226.31
	Two-Phase Region	304.81	
	Liquid Region	225.32	
LMTD (°C)	Gas Region	15.22	16.35
	Two-Phase Region	3.90	
	Liquid Region	3.91	
Heat Transfer Area (m²)	Gas Region	1.95	3.45
	Two-Phase Region	1.29	
	Liquid Region	1.37	

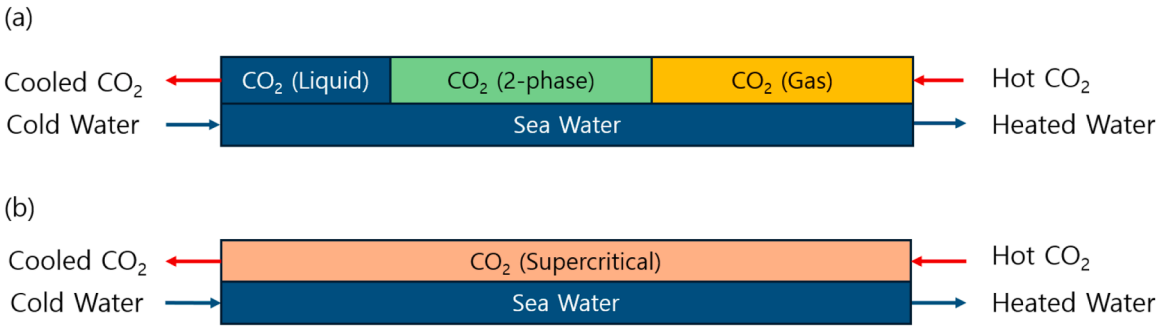


Fig. 7. Schematic diagram of the counter-flow heat exchanger under two different operating conditions: (a) Case 1 (Gas → Two-phase → Liquid), and (b) Case 2 (Supercritical → Supercritical).

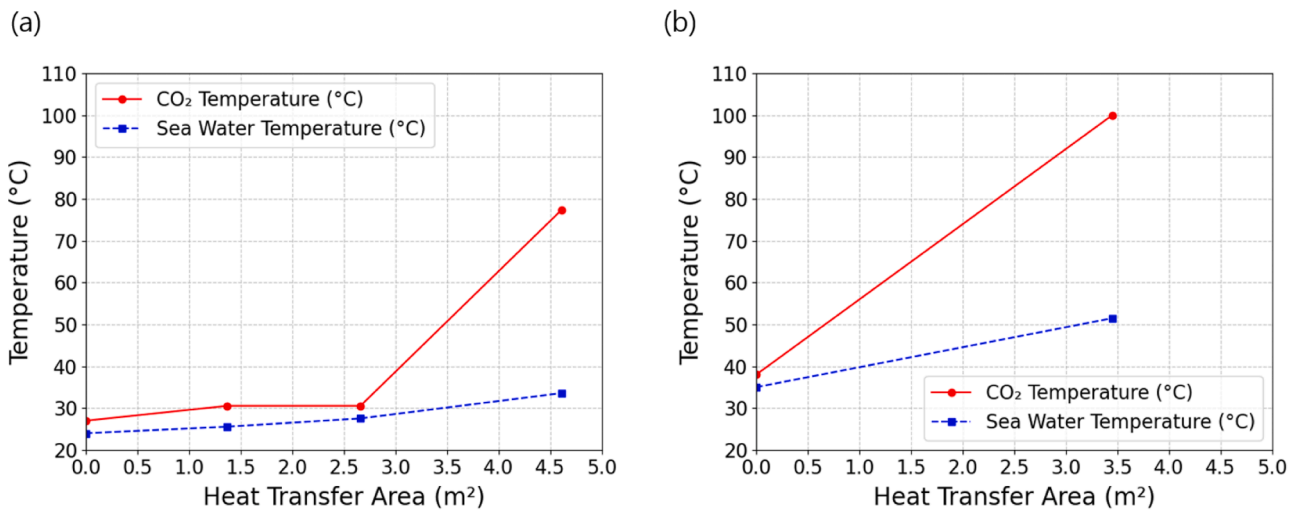


Fig. 8. Temperature distribution along the heat exchanger: (a) Case 1 (Gas → Two-phase → Liquid), and (b) Case 2 (Supercritical → Supercritical).

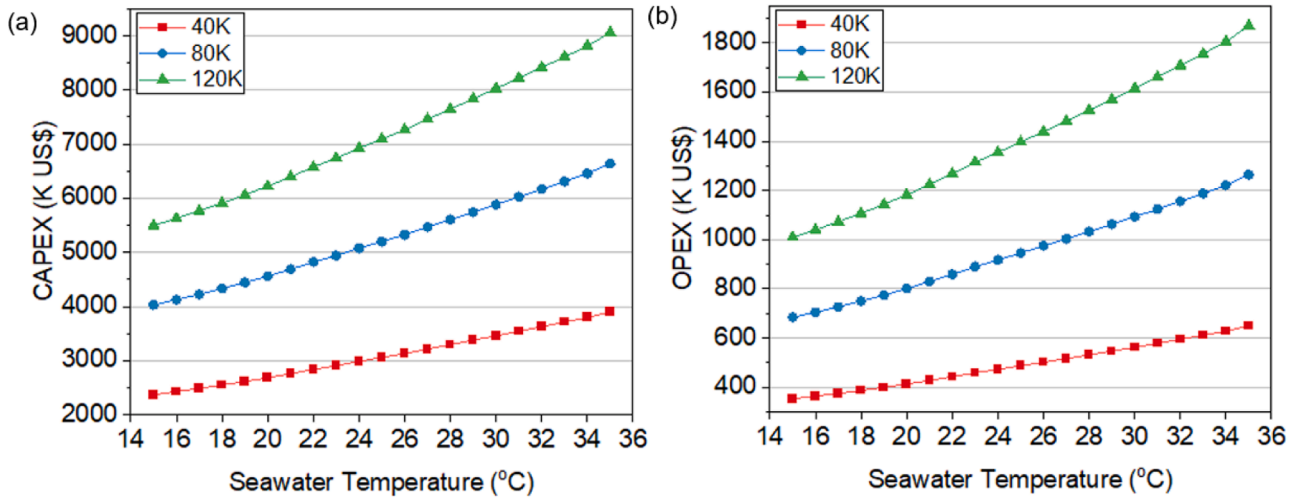


Fig. 9. Cost of reliquefaction as a function of the seawater temperature for (a) CAPEX and (b) OPEX.

temperatures rise, compressors need to handle increased power requirements, leading to higher CAPEX due to larger compressor capacities and increased OPEX resulting from higher energy consumption.

To ensure year-round operation, CAPEX must be estimated based on the highest seawater temperature of 34 °C, as sizing the equipment for average annual temperatures would be insufficient for peak summer conditions. On the other hand, OPEX is more reflective of average conditions, as operating during the summer incurs costs approximately 1.7 times higher than the annual average. For the cost analysis discussed in Section 4.2, CAPEX is estimated based on peak summer conditions, while OPEX is determined using the average seawater temperature of 17 °C.

4.2. CO_2 transport costs according to ship size, distance, and annual transport amount

Fig. 10 presents the transport costs for all 75 scenarios defined in Table 2. Fig. 10(a), 10(b), and 10(c) detail the transport cost per ton of CO_2 for ship sizes of 40 K, 80 K, and 120 K m^3, respectively, breaking down the costs by ship, reliquefaction, and terminal, further categorized into annualized CAPEX ($CAPEX_a$), fOPEX, and vOPEX. Additionally, Fig. 10 illustrates shipping efficiency, representing the percentage of time the ships are actively transporting CO_2 without idling during the 365 days of the year.

Because this study analyzed transport costs across a wide range of ship sizes and transport distances, the results showed a broad cost range, from a minimum of 5.49 USD/tCO_2 to a maximum of 86.64 USD/tCO_2. No single ship size was found to be universally advantageous across all transport distances and annual amounts. Additionally, it was found that larger ships become more economical as the transport distance and annual transport amount increase. For example, at the longest distance of 20,000 km and an annual CO_2 transport amount of 20 Mt, the transport cost for a 40 K ship was 68.56 USD/tCO_2, while for a 120 K ship, it was 49.62 USD/tCO_2, a reduction of 27.6 %. This confirms that larger ships are more cost-effective for long-distance, large-amount CO_2 transport.

For a transport distance of 1000 km, shipping efficiency was notably low, at 19 % for 1 $MtCO_2$/yr and 39 % for 2 $MtCO_2$/yr. This inefficiency is attributed to the oversized ship capacity relative to the transport distance and annual amount, making these scenarios economically unfeasible. The shipping efficiency increased to around 80 % as the transport distance lengthened and the annual transport amount grew. Therefore, optimizing ship size according to the annual amount and transport distance is crucial for planning CCS projects.

As Fig. 10 shows, the proportion of costs attributable to the ship increases noticeably as the transport distance grows. This trend is more

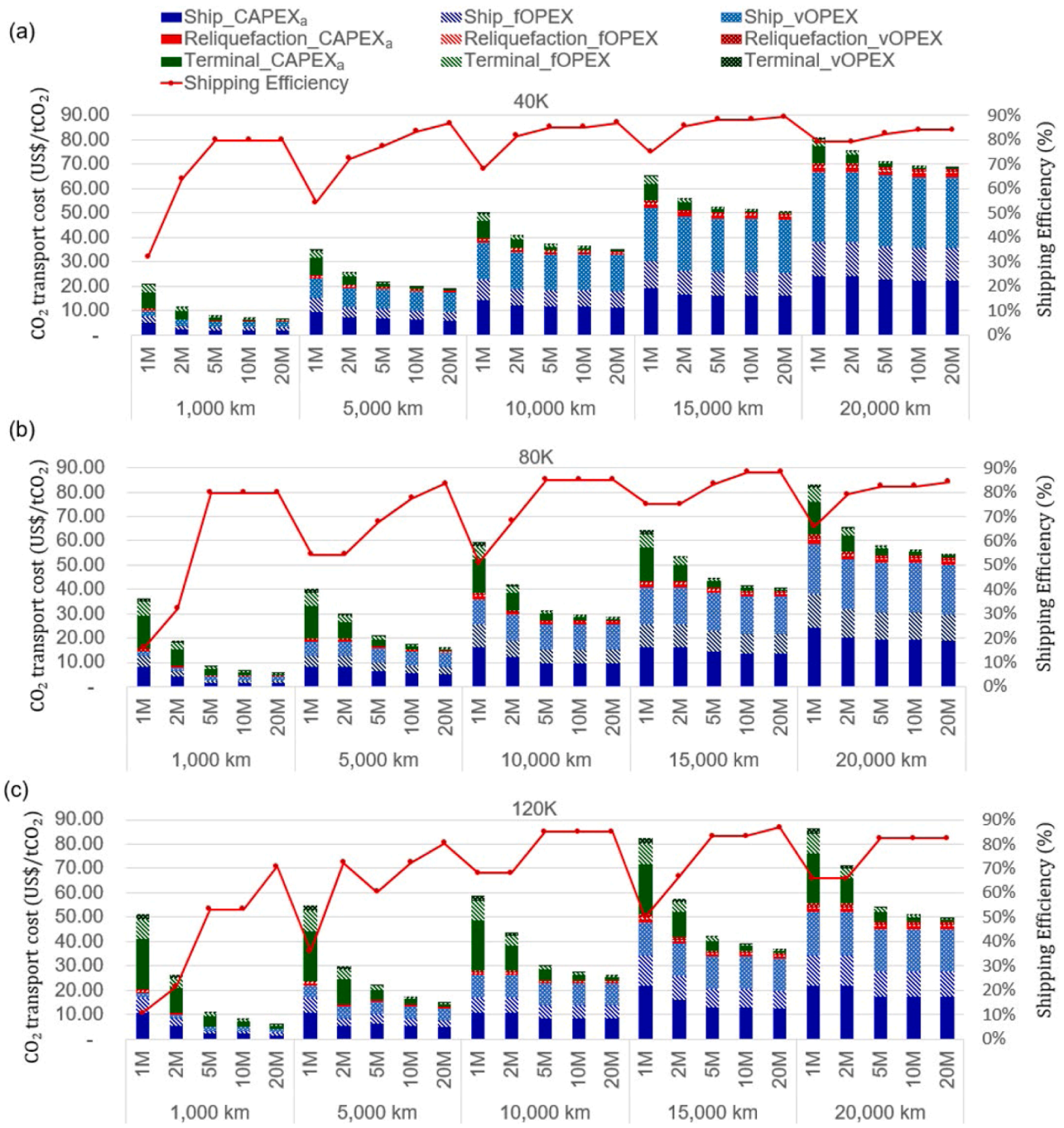


Fig. 10. CO_2 transport costs (US$/$tCO_2$) for (a) 40 K, (b) 80 K, and (c) 120 K ships, broken down by transportation distance and annual CO_2 transport amount. The bar charts represent the cumulative costs from various expenditure categories. The right Y-axis shows the shipping efficiency (%), depicted by the red line.

pronounced with smaller ships. As the annual transport amount increases, the cost per ton of CO_2 decreases, and the proportion of terminal costs in the total cost also decreases significantly. This trend is more evident with larger ships. To analyze these trends more closely, Fig. 11 presents the average proportions of $CAPEX_a$, fOPEX, and vOPEX in the total transport cost across various scenarios by ship size. Fig. 11 shows that smaller ships have a higher proportion of vOPEX costs, while the proportion of terminal $CAPEX_a$ and fOPEX is lower. This is because terminal costs are proportional to the size of a single ship but are independent of transport distance and annual transport amount.

Figs. 12 and 13 show transport costs based on changes in transport distance and annual transport amount. Data with a shipping efficiency below 50 % were excluded from this analysis as they were considered anomalies. Consequently, for an annual transport amount of 1 $MtCO_2$/yr with a transport distance of 1000 km, all data were excluded from Figs. 12 and 13. Fig. 12 illustrates the change in transport costs as transport distances vary, with the annual transport amount held constant and categorized by ship size. A proportional relationship between transport distance and transport cost is observed in Fig. 12. As the transport distance increases, the number of ships required also increases, as shown in Table 7, leading to a proportional rise in the overall costs for ships and reliquefaction systems. In other words, the $CAPEX_a$ and OPEX per ship remain constant, irrespective of transport distance or annual transport amount. Thus, transport costs increase in proportion to the number of ships required.

Fig. 13 illustrates the changes in transport costs as annual transport

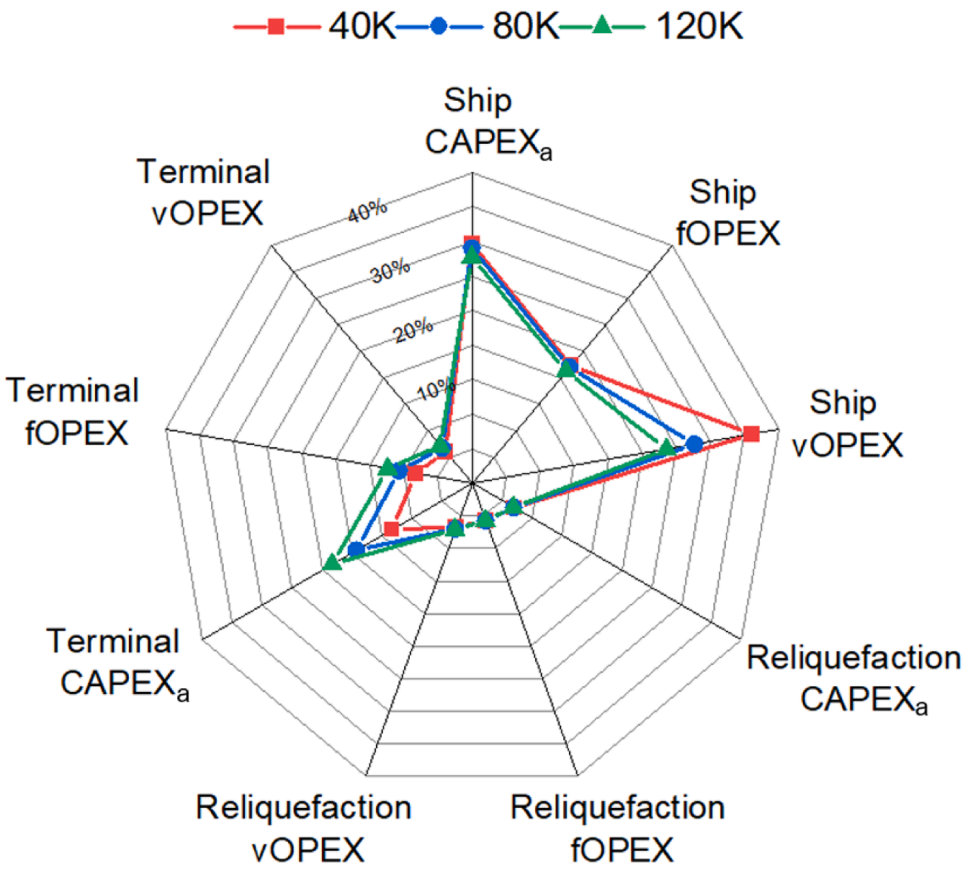


Fig. 11. Average impact of different cost categories on total transport costs, expressed as a percentage, for 40 K, 80 K, and 120 K ships across all scenarios.

amounts increase while keeping transport distance constant, categorized by ship size. Overall, as the annual transport amount increases, transport costs tend to decrease. This trend is particularly noticeable for larger ship sizes and longer transport distances. In Fig. 13(a)–13(e), the transport costs for 80 K and 120 K ships are generally lower than for 40 K ships, and this difference becomes more pronounced as the transport amount increases. However, the cost difference between 80 K and 120 K ships is smaller than between 40 K and 80 K ships. This suggests that while using ultra-large ships can reduce transport costs, the savings may not be as significant as expected. As shown in Table 5, the fuel consumption rate per unit of ship capacity does not decrease dramatically with larger ships.

Therefore, the key factor from a cost perspective is not just increasing ship size, but ensuring that the cost per ship and the fuel consumption rate per unit of cargo capacity decrease significantly.

4.3. Impact of reliquefaction on transport costs

This section explains the impact of BOG reliquefaction on transport costs. Fig. 10 shows the reliquefaction costs and their proportion of the total costs as a function of transport distance and annual transport amount. In Fig. 14, reliquefaction accounts for approximately 3,0 % to 6.0 % of the total transport costs, with an average of 4.6 %. This means that reliquefaction generally represents about 5 % of the overall CO_2 transport costs, which is not negligible in the context of total transport cost.

First, Fig. 14(a) and 14(b) display the cost per ton for reliquefaction and its percentage of the total transport cost according to the annual transport amount. As the annual transport amount increases, the proportion of reliquefaction costs in the total transport cost slightly increases, especially for larger ships. This occurs because, while the cost per ton for reliquefaction remains relatively steady or slightly decreases as the annual transport amount increases, the overall CO_2 transport costs decrease more significantly.

Fig. 14(c) and 14(d) show the cost per ton for reliquefaction and its percentage of the total transport cost according to the transport distance. As the transport distance increases, the cost per ton for reliquefaction rises noticeably. However, since the overall CO_2 transport costs also increase with longer distances, the proportion of reliquefaction costs only increases slightly with distance, with this trend being more pronounced for larger ships.

To provide a straightforward method for future studies to estimate reliquefaction costs without engaging in complex calculations, Table 12 presents the ratio of reliquefaction costs to ship $CAPEX_a$. These ratios, which range from approximately 15.1 % to 17.0 % of ship $CAPEX_a$, can serve as a useful reference for estimating costs in similar contexts. Comparing these results to a previous study by Jeon et al. (2016), which suggested that for a 12 K ship, reliquefaction $CAPEX_a$ and OPEX correspond to 2.5 % and 2.0 % of ship $CAPEX_a$ respectively, this study finds that reliquefaction $CAPEX_a$ and OPEX amount to higher percentages of ship $CAPEX_a$, specifically 6.9 % and 9.2 %.

4.4. Sensitivity analysis

We conducted a sensitivity analysis to determine the significance of factors influencing transport costs. The sensitivity analysis was performed using the scenario most likely to be commercially viable in the near future, which involved a 40 K ship size and a transport distance of 10,000 km. Although the 40 K ship is the smallest ship considered in this study, it is still the largest size among currently existing LCO_2 transport ships. This 10,000 km distance represents the route from Korea or Japan to potential CO_2 storage sites in Australia and the Middle East. These regions, particularly the industrially-focused East Asian countries, have high CO_2 emissions but limited CO_2 storage options, making the use of

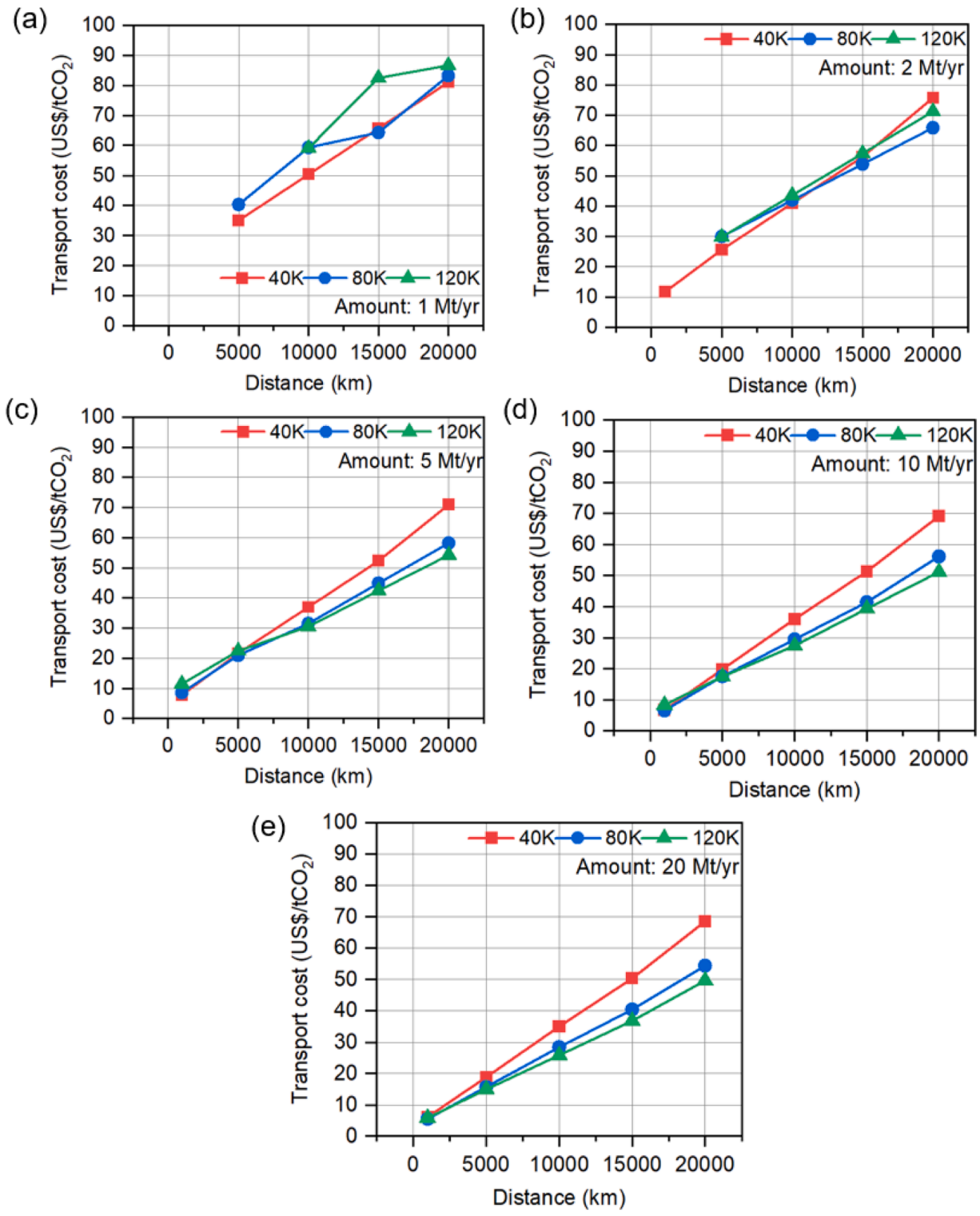


Fig. 12. CO_2 transport costs (US$/$tCO_2$) as a function of transport distance (km) for 40 K, 80 K, and 120 K ships at different annual transport amounts: (a) 2 $MtCO_2$/yr, (b) 5 $MtCO_2$/yr, (c) 10 $MtCO_2$/yr, and (d) 20 $MtCO_2$/yr.

storage sites in Australia or the Middle East highly probable.

Fig. 15 presents the results of the sensitivity analysis, showing the extent to which the factors considered in the study impact CO_2 transport costs. The fuel price used during ship operation had a significant impact on overall transport costs. A 20 % increase in fuel price resulted in a 9.65 % increase in total transport costs. The ship CAPEX also significantly affected transport costs, nearly as much as fuel price, with a 20 % increase leading to an 8.04 % rise in transport costs. The impact of these factors became more pronounced as the annual transport amount increased. While the discount rate was less impactful than the ship CAPEX, a 20 % increase still resulted in a 4.96 % rise in transport costs. Conversely, as the annual transport amount increased, the influence of the discount rate diminished. A 20 % increase in project lifetime led to a 2.33 % decrease in transport costs. The price of electricity had a minimal effect on CO_2 transport costs.

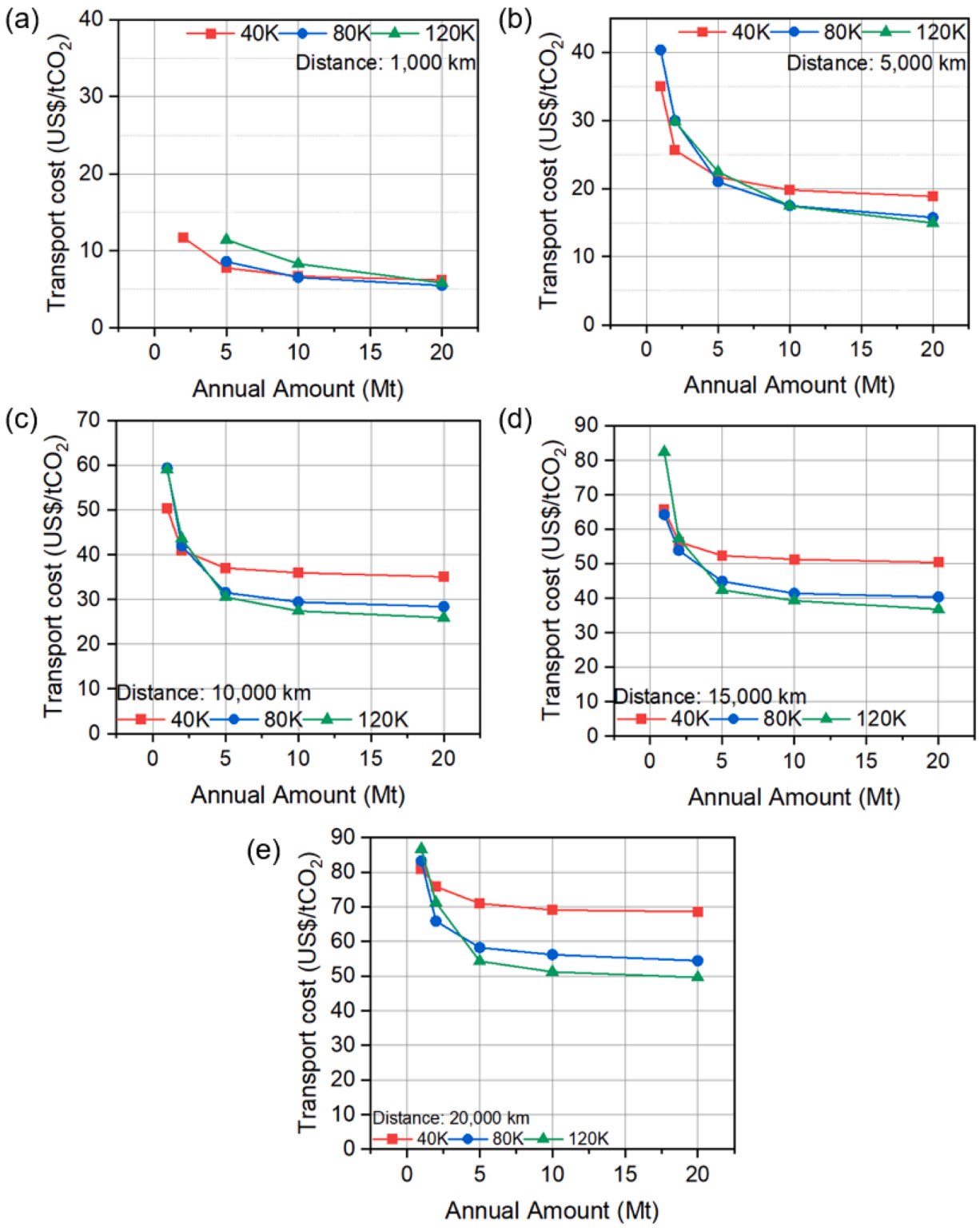


Fig. 13. CO_2 transport costs (US$/$tCO_2$) as a function of annual transport amount (Mt) for 40 K, 80 K, and 120 K ships across different transport distances: (a) 1000 km, (b) 5000 km, (c) 10,000 km, (d) 15,000 km, and (e) 20,000 km. Scenarios where the shipping efficiency is below 50 % are excluded from the data, as they are considered anomalies.

5. Conclusion

This study provides a comprehensive techno-economic analysis of CO_2 transport by ships over long distances, with a particular focus on evaluating costs related to BOG reliquefaction and the impact of varying seawater temperatures. The results show that while transport costs increase with distance, the cost per ton of CO_2 generally decreases as the annual transport amount increases, demonstrating economies of scale. However, there is no universally optimal ship size; the most cost-effective ship size depends on the specific transport distance and annual transport amount. Larger ships usually provide cost benefits in scenarios with large amounts and long distances, but the cost advantages

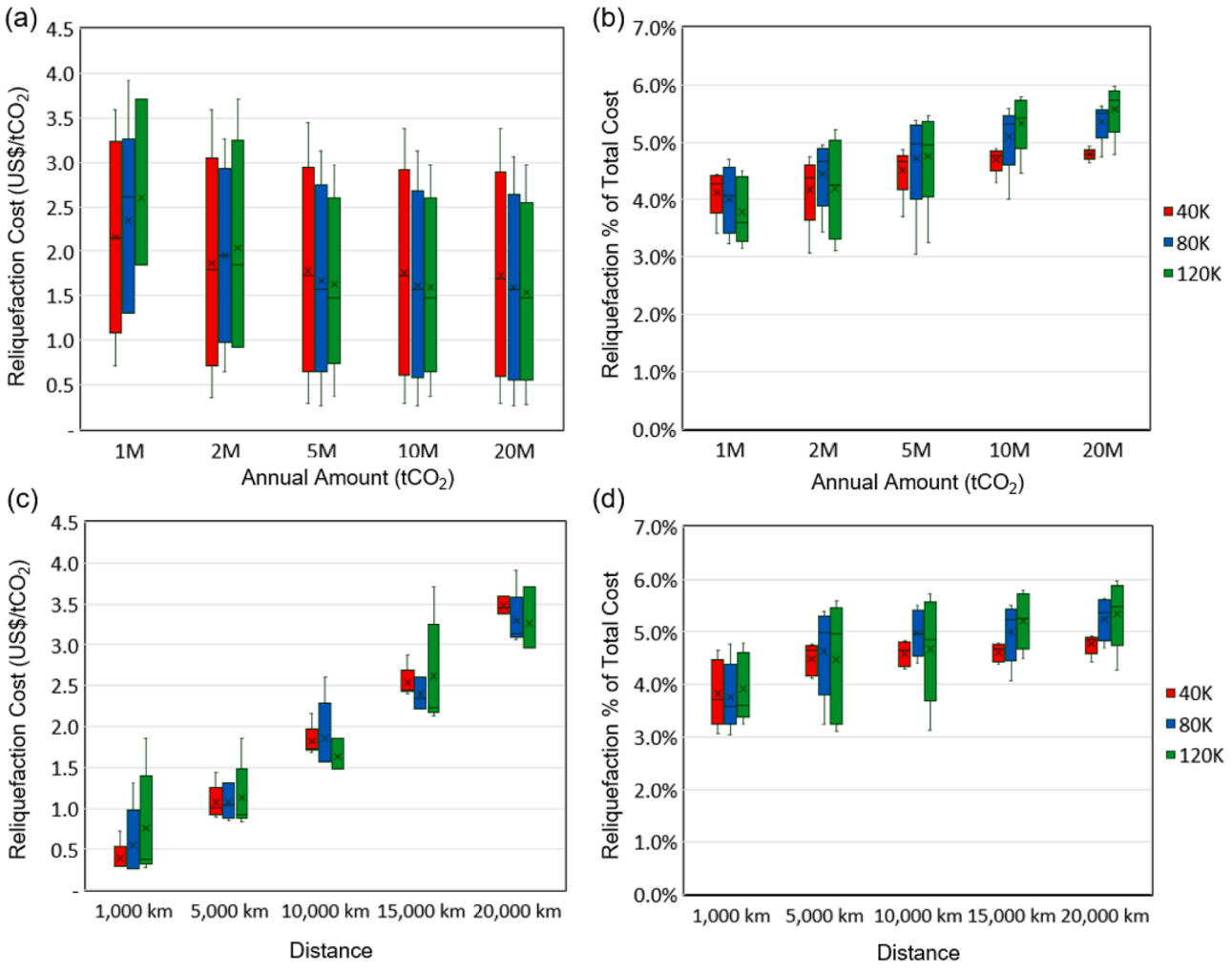


Fig. 14. Impact of reliquefaction on CO_2 transport costs, shown as (a) cost per ton and (b) percentage of total cost across different annual amounts, and (c) cost per ton and (d) percentage of total cost across different distances for 40 K, 80 K, and 120 K ships.

Table 12
Proportion of reliquefaction costs relative to Ship $CAPEX_a$.

Ship size (m^3)	Reliquefaction (% of Ship $CAPEX_a$)			
	$CAPEX_a$	fOPEX	vOPEX	Total
40K	6.8	2.5	5.8	15.1
80K	6.9	2.5	6.9	16.2
120K	6.9	2.5	7.6	17.0
Average	6.9	2.5	6.7	16.1

do not continue to grow significantly with further increases in ship size due to diminishing returns on fuel efficiency.

The study also demonstrates the significant impact of seawater temperature on the performance and cost of CO_2 reliquefaction. Higher seawater temperatures increase both the vapor ratio at the separator and the mass flow rate through the system, leading to increased compressor work and, consequently, higher CAPEX and OPEX. These effects are particularly pronounced at discharge temperature limits of 100 °C, emphasizing the importance of selecting appropriate discharge temperature limits to manage system efficiency.

Although BOG reliquefaction costs make up a small portion of total transport costs (about 3.0 % to 6.0 %), they are significant in long-distance, large-amount scenarios. The study found that reliquefaction costs contribute around 15.1 % to 17.0 % of the ship $CAPEX_a$, higher than previously reported. This finding highlights the importance of optimizing reliquefaction processes to maintain cost-effectiveness, especially as transport projects grow in size and distance.

The sensitivity analysis identified fuel prices and ship CAPEX as the most critical factors affecting total transport costs. These factors have a much larger impact than other variables, such as discount rates or electricity prices. This emphasizes the need to focus on optimizing ship design and managing costs to ensure that CO_2 transport by ship remains economically viable, particularly for large-scale, long-distance CCS projects.

In summary, the study underscores the need for careful planning and optimization of ship-based CO_2 transport systems, especially in managing ship CAPEX and fuel costs, to improve the economic feasibility of long-distance CCS projects. Future research should explore advanced optimization techniques, alternative reliquefaction schemes such as mixed refrigerant cycles or direct expansion systems, and new technologies to further reduce costs and enhance efficiency.

CRediT authorship contribution statement

Hyonjeong Noh: Writing – original draft, Visualization, Resources, Methodology, Data curation. **Kwangu Kang:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

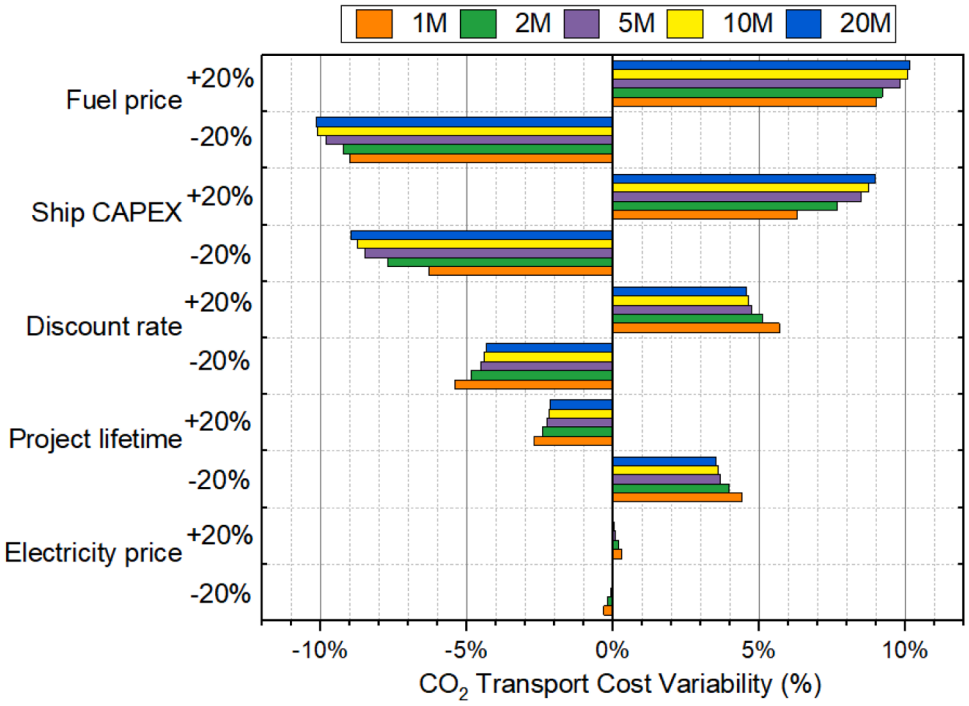


Fig. 15. Sensitivity analysis results showing the variability in CO_2 transport costs (%) in response to changes in key factors: fuel price, ship CAPEX, discount rate, project lifetime, and electricity price. Each factor's impact is analyzed with $a \pm 20$ % variation across different annual CO_2 transport amounts (1, 2, 5, 10, and 20 $MtCO_2/yr$).

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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Data availability

Data will be made available on request.

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